

Genetic report

Psychedelics

Sérgio F · Report date: 02-10-2026

Categories: **MENTAL HEALTH · DETOX**

This report repeats, with DNA of Sérgio F, the study of an example report on psychedelics (22 traits, the same rsIDs and equivalent methods). It is informative and educational. It is not a diagnosis or prescription and does not encourage the use of controlled substances. The use of psychedelics outside the clinical context or authorized trial can be illegal and has risks. All recommendations are marked as a suggestion (not applied) and should be discussed with a doctor. Polygenic scores explain a small part of the risk; the absolute risk of each condition remains low in most people.

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Summary

22

traits

330

rsIDs of the
example report

170

in the chip

142

imputed
(DR2≥0.8)

18

SNP not found in
the data

Of the unique 330 rsIDs of the example report (349 lines, because some SNPs enter several traits), 170 come directly from the genotyping chip and 142 from the imputation with quality DR2≥0.8. 18 were not found: 16 were imputed with low quality (DR2<0.8, treated as not found) and the remaining are indels or do not exist in the data. Extra metabolism and context 61 SNPs (CYP3A5, CYP1A2, UGT1A9/1A1, MAOA, COMT, OPRM1, DRD2, OXTR, QT SNPs, among others) were also analyzed. Polygenic scores (PGS Catalog) used 5 220 362 variants in total.

Main results

Trait	Result	Interpretation
Schizophrenia (PGS)	MORE LIKELY	More likely to have schizophrenia
Bipolar disease (PGS)	MORE LIKELY	More likely to have bipolar disease
Borderline disorder (small GWAS)	MORE LIKELY	More likely to have borderline disturbance
BDNF Val66Met	SLIGHTLY LOWER LEVELS	Relatively lower BDNF levels (1 Met allele)
FAAH C385A	LOWER ACTIVITY	FAAH activity probably lower (1 allele A)
NRG1	LOWER ACTIVITY	Predisposition for lower NRG1 activity
Effects of THC	INCREASED	Predisposition for increased psychedelic effects
CYP2C9	INTERMEDIATE METABOLIZER	Intermediate metabolizer (*1/*3; activity score 1.0)
CYP2B6	HIGHER ACTIVITY	Normal to increased activity (probable *2/*4; fast metabolizer)

What weighs the most in practice

- 1) The PGS of schizophrenia (p88) and bipolar disease (p79) are above average. They are not diagnostic, but they are the type of signal that clinical trials with psychedelics deal with prudence (family history remains the strongest criterion).
- 2) The lithium stack (5 mg) has a sign of seizures with classic psychedelics in reports (Nayak 2021), with therapeutic doses; this weighs more than any SNP.
- 3) CYP2C9 *1/*3 (intermediate): exposure to THC probably higher. CYP2B6 *2/*4: ketamine should not be exposed anymore (without *6).
- 4) Ibogain: clinical cardiac risk (QT) regardless of genetics; the evaluated QT SNPs do not show risk alleles, but this does not protect.

General view of the results

Colors: **above/below the typical** **typical** **favourable** **activity of genes (no judgment good/bad)**
not reproducible. The labels in English keep the categories of the example report.

Psychedelics Response

BDNF

SLIGHTLY LOWER LEVELS

Relatively lower BDNF levels (1 Met allele)

Psychedelic Effects of THC (Functional)

INCREASED

Predisposition for increased psychedelic effects

HTR2A (Serotonin)

TYPICAL ACTIVITY

HTR2A activity probably typical; without alleles linked to more receptors (reference genome)

CNR1 (Cannabinoids)

TYPICAL ACTIVITY

CNR1 activity probably typical

Response to Ketamine

TYPICAL

Predisposition for typical ketamine response

Psychedelics Metabolism

CYP2D6 (Mental Health, Detox)

TYPICAL ACTIVITY

Probable typical activity (*1/*2; probable normal metabolizer, without CNV evaluated)

CYP2B6 (Drug Metabolism)

HIGHER ACTIVITY

Normal to increased activity (probable *2/*4; fast metabolizer)

CYP2C9 (Detox)

INTERMEDIATE METABOLIZER

Intermediate metabolizer (*1/*3; activity score 1.0)

CYP2C19 (Detox)

EXTENSIVE METABOLIZER

Probable extensive/normal metabolizer (*1/*1)

CYP3A4 (Detox)

TYPICAL ACTIVITY

Probable typical CYP3A4 activity (*1/*1, without *22)

FAAH (Cannabinoids)

LOWER ACTIVITY

FAAH activity probably lower (1 allele A)

Mental Health Risk

Bipolar Disorder

MORE LIKELY

More likely to have bipolar disease

THC and Psychosis

TYPICAL LIKELIHOOD

Typical probability of psychotic symptoms with cannabis

Borderline Personality

MORE LIKELY

More likely to have borderline disturbance

Psychological Trauma

TYPICAL LIKELIHOOD

Typical probability of PTSD

Non-Epileptic Seizures

NOT REPRODUCIBLE

No published genetic data to reproduce this trait (proxy: neuroticism)

Schizophrenia

MORE LIKELY

More likely to have schizophrenia

Anxiety

TYPICAL LIKELIHOOD

Typical anxiety probability

Panic Attacks

TYPICAL LIKELIHOOD

Typical probability of panic attacks

Seizures

TYPICAL LIKELIHOOD

Typical probability of seizures (epilepsy)

Mental Health Genes

DISC1 (Mental Health)

TYPICAL ACTIVITY

Predisposition for typical DISC1 activity

NRG1 (Mental Health)

LOWER ACTIVITY

Predisposition for lower NRG1 activity

Overview of the recommendations

The 50 recommendations of the example report were reordered by Sérgio F results: the risks with a result above the typical (bipolar, borderline, BDNF) weigh more and the typical ones weigh less. Items with conflict with the stack descend. *All: suggestion (not applied).*

#	Recommendation	Dose (example)	Targeted risks	Priority	State in context
1	Psychotherapy Psychotherapy	1 hour	Bipolar Disorder ↑, Psychological Trauma	33,8	compatible with stack
2	Cognitive behavioral therapy (TCC) Cognitive-Behavioral Therapy (CBT)	—	Bipolar Disorder ↑, Borderline Personality ↑, Psychological Trauma	17,8	compatible with stack
3	Omega-3 (fish oil) Omega-3 (Fish Oil)	500 mg	Bipolar Disorder ↑, BDNF ↑, Borderline Personality ↑, Psychological Trauma	15,4	in the stack
4	Solar exposure Sunlight Exposure	20 minutes	Bipolar Disorder ↑	12,0	caution (interaction or context)
5	Vitamin C Vitamin C	500 mg	Bipolar Disorder ↑	9,0	in the stack
6	Yoga Yoga	30 minutes	Bipolar Disorder ↑, BDNF ↑, Borderline Personality ↑, Psychological Trauma	8,1	compatible with stack
7	Dance Dance	30 minutes	Borderline Personality ↑, Psychological Trauma	7,1	compatible with stack
8	DHA (Omega-3) DHA (Omega-3)	200 mg	BDNF ↑, Psychological Trauma	5,4	in the stack
9	Modification of attentional bias Attention Bias Modification	—	Psychological Trauma	4,2	compatible with stack
10	Mindfulness based cognitive therapy (MBCT) Mindfulness-Based Cognitive Therapy (MBCT)	2 hours	Psychological Trauma	4,2	compatible with stack
11	Meditation Meditation	30 minutes	Bipolar Disorder ↑, Borderline Personality ↑, Psychological Trauma	4,1	compatible with stack
12	Mindfulness Mindfulness	30 minutes	Borderline Personality ↑, Psychological Trauma, Non-Epileptic Seizures	3,3	compatible with stack
13	Mindfulness-based stress reduction (MBSR) Mindfulness-Based Stress Reduction (MBSR)	2 hours	Borderline Personality ↑, Psychological Trauma	3,1	compatible with stack
14	Biofeedback Biofeedback	—	Psychological Trauma	3,1	compatible with stack
15	Strength training Strength Training	48 hours	Psychological Trauma	3,1	in the stack
16	Art therapy Art Therapy	1 hour	Bipolar Disorder ↑, Borderline Personality ↑, Psychological Trauma	2,7	compatible with stack
17	Animal assisted therapy Pet Therapy	30 minutes	Psychological Trauma	2,1	compatible with stack
18	EFT (tapping) EFT Tapping	20 minutes	Psychological Trauma	2,1	compatible with stack
19	Transcendental meditation Transcendental Meditation	20 minutes	Psychological Trauma	2,1	compatible with stack
20	Relaxation applied Applied Relaxation	30 minutes	Psychological Trauma	1,4	compatible with stack
21	Acceptance therapy and commitment (ACT)	—	Bipolar Disorder ↑, Borderline Personality ↑, Psychological Trauma	1,4	compatible with stack

22	Omega-3 food Dietary Omega-3 Fatty Acids	—	Bipolar Disorder ↑, Borderline Personality ↑	1,0	in the stack
23	Black chocolate Dark Chocolate	—	Bipolar Disorder ↑, BDNF ↑	1,0	caution (interaction or context)
24	Lithium orate Lithium Orote	5 mg	Bipolar Disorder ↑	1,0	in the stack
25	Dry fruits Nuts	3 oz	Bipolar Disorder ↑	1,0	compatible with stack
26	Whole grains Whole Grains	—	Bipolar Disorder ↑	1,0	compatible with stack
27	Scaled Climbing	1 hour	Psychological Trauma	0,7	compatible with stack
28	Tryptophan Tryptophan	500 mg	Bipolar Disorder ↑	0,4	not recommended in this context
29	Music therapy Music Therapy	30 minutes	Psychological Trauma	0,3	compatible with stack
30	Lavender essential oil Lavender Essential Oil	2 drop	Psychological Trauma	0,3	compatible with stack
31	Diaphragmatic breathing Diaphragmatic Breathing	10 minutes	Psychological Trauma	0,3	compatible with stack
32	Kundalini yoga Kundalini Yoga	1 hour	Psychological Trauma	0,3	compatible with stack
33	Intensive light therapy in the morning Morning Bright Light Therapy	20 minutes	Psychological Trauma	0,3	caution (interaction or context)
34	N-back training N-Back Training	20 minutes	Psychological Trauma	0,3	compatible with stack
35	Mindfulness meditation Mindfulness Meditation	30 minutes	BDNF ↑	0,0	compatible with stack
36	Ginkgo Ginkgo	120 mg	BDNF ↑	0,0	caution (interaction or context)
37	Curcumin Curcumin	500 mg	BDNF ↑	0,0	not recommended in this context
38	D-manose D-Mannose	2g	Bipolar Disorder ↑	0,0	not recommended in this context
39	Bacopa Bacopa	300 mg	BDNF ↑	0,0	caution (interaction or context)
40	Cocoa Cocoa	—	BDNF ↑	0,0	caution (interaction or context)
41	Green tea (extract) Green Tea	400 mg	BDNF ↑	0,0	not recommended in this context
42	Time in nature Spend Time in Nature	2 hours	BDNF ↑	0,0	compatible with stack
43	Resilient starch Resistance Starch	40 g Hi-Maize	BDNF ↑	0,0	caution (interaction or context)
44	Progressive muscle relaxation Progressive Muscle Relaxation	10 minutes		0,0	compatible with stack
45	Food zinc Dietary Zinc	—		0,0	in the stack
46	Galphimia Galphimia	350 mg		0,0	caution (interaction or context)
47	Nidra Nidra Yoga	30 minutes	Psychological Trauma	0,0	compatible with stack
48		10 mg		0,0	

	CBD CBD			not recommended in this context
49	Rhyke therapy Last Therapy	30 minutes	0,0	compatible with stack
50	Massage Massage	30 minutes	0,0	compatible with stack

↑ = trait with a result above the typical in Sérgio F. Priority = $\Sigma(\text{impact} \times \text{evidence} \times \text{result factor})$. The 10 points of risk reduction are in its own section.

Psychedelics Response: results in detail

Traits of gene and neurotransmitter activity: no good/bad judgment.

BDNF

SLIGHTLY LOWER LEVELS Relatively lower BDNF levels (1 Met allele)

Score of 1 of 1 variants (sum of effect alleles). In the reference population (n=633): the 81st percentile (mean position); 32% have exactly the same score.

The BDNF gene produces the brain-derived neurotrophic factor, central to neuronal plasticity. The rs6265 variant (Val66Met) alters the transport and release of BDNF; the T allele (Met) is associated with lower activity-dependent release. In some studies, Met was associated with a lower antidepressant effect of ketamine at low dose, but not consistently in resistant depression. The cognitive effects described come mainly from clinical samples and are inconsistent.

Effect allele (in the sense of result) (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
BDNF	rs6265	CT	chip	T	1	Met: less BDNF released

Psychedelic Effects of THC (Functional)

INCREASED Predisposition for increased psychedelic effects

Score of 7 of 7 variants (sum of effect alleles). In the reference population (n=633): the 79th percentile (mean position); 16% have exactly the same score.

7 score variants of the endocannabinoid and dopaminergic system (CNR1, FAAH, COMT, DRD2, AKT1), with the directions described in the example report: CNR1 rs1049353 T and rs12720071 T (lower CB1 expression; more psychotic symptoms in consumers), FAAH rs324420 A (less FAAH, plus anandamide) and rs2295633 A (G is associated with more FAAH), COMT rs4680 G (Val), DRD2 rs1076560 A and AKT1 rs2494732 C.

Effect allele (in the sense of result) (5)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
CNR1	rs1049353	CT	chip	T	1	less CB1 expression; more psychotic symptoms with cannabis
FAAH	rs2295633	AG	chip	A	1	G = more FAAH (protector); A : sense of risk
COMT	rs4680	GG	chip	G	2	Val: plus COMT; risk of psychosis with cannabis
CNR1	rs12720071	TT	chip	T	2	psychotic symptoms with cannabis
FAAH	rs324420	CA	chip	A	1	less FAAH; more anandamide

No effect allele (meaning contrary / reference) (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
TTC12	rs1076560	CC	chip	A	0	plus D2 signaling; psychosis with cannabis
AKT1	rs2494732	TT	chip	C	0	Psychosis in daily cannabis smokers

HTR2A (Serotonin)

TYPICAL ACTIVITY HTR2A activity probably typical; without alleles linked to more receptors (reference genome)

Score of 2 of 2 variants (sum of effect alleles). In the reference population (n=633): the 16th percentile (mean position); 33% have exactly the same score.

The 5-HT2A receptor is the main target of classical psychedelics (psilocin, LSD, DMT, mescaline). The minor alleles rs6313 A (T102C) and rs6311 T (−1438G>A), inherited together, are associated with more active receptors in some studies. We added rs6314 (His452Tyr), which is not in the sample: in vitro, the variant Tyr452 modifies in a modest and specific way of each drug the signaling of psilocin, 5-MeO-DMT and LSD (Schmitz 2022).

No effect allele (meaning contrary / reference) (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
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HTR2A	rs6313	GG	chip	A	0	more active 5-HT2A receptors
HTR2A	rs6311	CC	chip	T	0	more active 5-HT2A receptors

CNR1 (Cannabinoids)

TYPICAL ACTIVITY CNR1 activity probably typical

7 score of 7 variants (sum of effect alleles with signal). In the reference population (n=633): the 34th percentile (mean position); 26% have exactly the same score.

CNR1 encodes the cannabinoid receptor CB1. Score with 7 signal variants, with the directions of the example report: less activity/expression with rs1049353 T, rs6454674 G, rs806380 G, rs806374 C and rs2180619 G; plus with rs2023239 C and rs806378 T.

Variants that pull for more activity (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
CNR1	rs806378	CT	chip	T	1	plus CNR1 activity

Variants that pull for less activity (meaning opposite) (4)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
CNR1	rs1049353	CT	chip	T	1	less CB1 expression; more psychotic symptoms with cannabis
CNR1	rs806380	AG	chip	G	1	less CB1
CNR1	rs806374	TC	chip	C	1	less CB1 (dentestatin, fat)
CNR1	rs6454674	TG	chip	G	1	less CB1

No effect allele (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
CNR1	rs2180619	AA	chip	G	0	less CB1 in the brain
CNR1	rs2023239	TT	imputation DR2=0.99	C	0	plus CB1

Response to Ketamine

TYPICAL Predisposition for typical ketamine response

Signal score (7 of 7 variants with published effect allele). Mean position: the 12th percentile. Convention of uncertain signal and small GWAS, without replication: labeled TYPICAL.

The 13 variants of the example report come from two small GWAS of antidepressant response to ketamine/sketamine (Guo 2018; Li 2020), without replication. Several are rare and, for three of them, the catalog does not indicate the effect allele. The factors with the most influence on the response are the dose, the route, the treated disease and the concomitant drugs.

With allele associated with more response (5)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
GGH	rs4739050	GA	imputation DR2=1.00	A	1	associated with increased response score (Guo 2018/Li 2020)
SLC22A15	rs79749176	TT	imputation DR2=1.00	T	2	associated with increased response score (Guo 2018/Li 2020)
SLC22A15	rs6689906	CC	imputation DR2=0.97	C	2	associated with increased response score (Guo 2018/Li 2020)
SEC11A	rs55945116	GC	imputation DR2=0.99	G	1	associated with increased response score (Guo 2018/Li 2020)
DUSP18	rs5997786	CC	imputation DR2=0.82	C	2	associated with increased response score (Guo 2018/Li 2020)

With allele associated with less response (meaning opposite) (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
FAM83B	rs112647602	GG		G	2	

			imputation DR2=0.89			associated with decreased response score (Guo 2018/Li 2020)
—	rs1846786	TT	imputation DR2=0.99	T	2	associated with decreased response score (Guo 2018/Li 2020)

No effect or unpublished effect (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
TMBIM4	rs17767394	CC	imputation DR2=0.93	—	—	Allele of unpublished effect
ZBTB14	rs75908125	GC	imputation DR2=0.98	—	—	Allele of unpublished effect

SNP not found in the data (4)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
IRAK3	rs11465988	—	SNP not found in the data (Imputation DR2=0.61, below the threshold 0.8)	—	—	Allele of unpublished effect
TGFA	rs115141868	—	SNP not found in the data (Imputation DR2=0.42, below the threshold 0.8)	—	—	Allele of unpublished effect
CXXC4	rs151184257	—	SNP not found in the data (Imputation DR2=0.34, below the threshold 0.8)	—	—	Allele of unpublished effect
CRIM1	rs1524145	—	SNP not found in the data (Imputation DR2=0.69, below the threshold 0.8)	G	—	associated with decreased response score (Guo 2018/Li 2020)

Psychedelics Metabolism: results in detail

CYP2D6 (Mental Health, Detox)

TYPICAL ACTIVITY Probable typical activity (*1/*2; probable normal metabolizer, without CNV evaluated)

Diplotype inferred from chip SNPs and imputed (CNV and non-visible indels).

CYP2D6 has more than 100 alleles and may be deleted or duplicated (CNV). We tested *2, *3, *4, *6, *7, *8, *9, *10, *17, *29 and *41. It is the enzyme with the most known impact on LSD (slow metabolizers have about +75% of exposure and longer effects), MDMA, 5-MeO-DMT, ibogain and ayahuasca harmaline.

SNPs used in the diploid

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
CYP2D6	rs16947	AG	chip	—	—	*2 (R296C), normal function
CYP2D6	rs1065852	GG	chip	A	0	*10 (P34S), reduced function
CYP2D6	rs3892097	CC	chip	T	0	*4 (splicing), without function
CYP2D6	rs5030655	—	SNP not found in the data	—	—	*6 (del T), without function
CYP2D6	rs35742686	—	SNP not found in the data	—	—	*3 (del A), without function
CYP2D6	rs5030867	TT	chip	G	0	*7, without function
CYP2D6	rs5030865	CC	chip	—	—	*8/*14, without function/reduced
CYP2D6	rs5030656	—	SNP not found in the data	—	—	*9 (del K281), reduced
CYP2D6	rs28371706	—	SNP not found in the data (Imputation DR2=0.00, below the threshold 0.8)	A	—	*17 (T107I), reduced
CYP2D6	rs28371725	CC	chip	T	0	*41 (splicing), reduced
CYP2D6	rs1135840	GC	chip	—	—	S486T (*2/*10 marker)
CYP2D6	rs1080985	CG	chip	—	—	-1584C>G (*2 marker)

Undefined effect allele (4)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
CYP2D6	rs5030867	TT	chip	—	—	
CYP2D6	rs3892097	CC	chip	—	—	
CYP2D6	rs1065852	GG	chip	—	—	
CYP2D6	rs16947	AG	chip	—	—	

SNP not found in the data (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
CYP2D6	rs5030655	—	SNP not found in the data	—	—	
CYP2D6	rs28371706	—	SNP not found in the data (Imputation DR2=0.00, below the threshold 0.8)	—	—	

CYP2B6 (Drug Metabolism)

HIGHER ACTIVITY Normal to increased activity (probable *2/*4; fast metabolizer)

Diplotype inferred from chip SNPs and imputed (CNV and non-visible indels).

CYP2B6 is the main pathway of ketamine N-demethylation and participates in MDMA. *6 (rs3745274 T + rs2279343 G) reduces depuration; *4 (only rs2279343 G) is associated with equal or increased activity; *2 (rs8192709 T) has normal function.

SNPs used in the diploid

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
CYP2B6	rs3745274	GG	chip	T	0	*6/*9 (Q172H), reduced
CYP2B6	rs2279343	AG	chip	G	1	*4/*6 (K262R)
CYP2B6	rs28399499	TT	chip	C	0	*18 (I328T), without function
CYP2B6	rs8192709	CT	chip	T	1	*2 (R22C)
CYP2B6	rs3211371	—	SNP not found in the data (Imputation DR2=0.00, below the threshold 0.8)	T	—	*5 (R487C)
CYP2B6	rs4803419	CT	chip	T	1	Intronica 15582C>T (lower expression)

Undefined effect allele (4)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
CYP2B6	rs3745274	GG	chip	—	—	
—	rs2279343	AG	chip	—	—	
CYP2B6	rs28399499	TT	chip	—	—	
CYP2B6	rs8192709	CT	chip	—	—	

SNP not found in the data (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
CYP2B6	rs3211371	—	SNP not found in the data (Imputation DR2=0.00, below the threshold 0.8)	—	—	

CYP2C9 (Detox)

INTERMEDIATE METABOLIZER Intermediate metabolizer (*1/*3; activity score 1.0)

Diploype inferred from chip SNPs and imputed (CNV and non-visible indels).

CYP2C9 transforms THC into 11-OH-THC. *3/*3 carriers had about 3 times more exposure to THC (Sachse-Seeboth 2009). It also participates in the LSD in vitro, without effect demonstrated in vivo.

SNPs used in the diploid

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
CYP2C9	rs1799853	CC	chip	T	0	*2 (R144C), reduced
CYP2C9	rs1057910	AC	chip	C	1	*3 (I359L), reduced
CYP2C9	rs28371686	CC	chip	G	0	*5, reduced
CYP2C9	rs7900194	GG	chip	A	0	*8, reduced
CYP2C9	rs28371685	CC	chip	T	0	*11, reduced
CYP2C9	rs9332131	—	SNP not found in the data	—	—	*6 (del A), without function

Undefined effect allele (5)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
CYP2C9	rs1057910	AC	chip	—	—	
CYP2C9	rs7900194	GG	chip	—	—	
CYP2C9	rs28371686	CC	chip	—	—	
CYP2C9	rs28371685	CC	chip	—	—	
CYP2C9	rs1799853	CC	chip	—	—	

CYP2C19 (Detox)

EXTENSIVE METABOLIZER Probable extensive/normal metabolizer (*1/*1)

Diplotype inferred from chip SNPs and imputed (CNV and non-visible indels).

CYP2C19 metabolizes CBD and participates in MDMA. *2 and *3 have no function; *17 increases expression.

SNPs used in the diploid

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
CYP2C19	rs4244285	GG	chip	A	0	*2, without function
CYP2C19	rs4986893	GG	chip	A	0	*3, without function
CYP2C19	rs28399504	AA	chip	G	0	*4, without function
CYP2C19	rs12248560	CC	chip	T	0	*17, increased function

Undefined effect allele (3)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
NOC3L	rs12248560	CC	chip	—	—	
CYP2C19	rs4986893	GG	chip	—	—	
CYP2C19	rs4244285	GG	chip	—	—	

CYP3A4 (Detox)

TYPICAL ACTIVITY Probable typical CYP3A4 activity (*1/*1, without *22)

Diplotype inferred from chip SNPs and imputed (CNV and non-visible indels).

CYP3A4 participates in ketamine, LSD (nor-LSD), psilocin (minor), THC and CBD. *22 reduces the activity to about half. CYP3A inhibitors (ketoconazole, fluconazole, diltiazem, grapefruit, berberine and some polyphenols) can increase the levels of these substances and rapamycin.

SNPs used in the diploid

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
CYP3A4	rs35599367	GG	chip	A	0	*22, reduced
CYP3A5	rs776746	CT	chip	C	1	*3 (splicing): nonexpressor

Undefined effect allele (7)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
CYP3A4	rs28371759	AA	chip	—	—	
CYP3A43	rs680055	CC	chip	—	—	
CYP3A4	rs55785340	AA	chip	—	—	
TRIM4	rs35599367	GG	chip	—	—	
TRIM4	rs2740574	TT	chip	—	—	
CYP3A5	rs2242480	CT	chip	—	—	
—	rs12721627	GG	chip	—	—	

FAAH (Cannabinoids)

LOWER ACTIVITY FAAH activity probably lower (1 allele A)

Score of 1 of 1 variants (sum of effect alleles). In the reference population (n=633): the 80th percentile (mean position); 33% have exactly the same score.

FAAH degrades anandamide. The rs324420 A allele (P129T) reduces the stability of the enzyme and rises the anandamide. Described with less anxiety in acute stress and less PTSD, but also with more problematic substance consumption.

Effect allele (in the sense of result) (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
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Metabolism by substance

Joint reading of the metabolism pathways of each substance with Sérgio F genotypes (including extra SNPs). Pharmacological information, no indication of use.

Substance	Main routes	Sérgio F Genotype	Implication	Evidence
Psilocybin / psilocin	Phosphatases → psilocin; glucuronidation by UGT1A10 (intestine) and UGT1A9 (liver), the main pathway; MAO-A → 4-HIAA; CYP2D6 and CYP3A4 minor.	UGT1A9 without *3 (rs72551330 T/T), –2152 C/C, –275 T/T: typical function. UGT1A10 not evaluated. MAOA rs6323 T (hemizigote; marker associated with lower activity in some studies; uVNTR is not visible). Probable normal CYP2D6.	Typical expected exposure. In one study, the CYP2D6 genotype did not alter free psilocin (Thomann 2024). Theoretically, UGT/MAO inhibitors can increase exposure.	moderate (via) · weak (genotype)
LSD	CYP2D6 (main determinant in vivo); CYP1A2, CYP2C9, CYP2C19 and CYP3A4 in vitro.	Probable CYP2D6 *1/*2 (without *3/*4/*6/*9/*10/*41); CNV not evaluated.	Typical exposure and duration, if there is no deletion or duplication. Slow metabolizers have about +75% of exposure and longer effects (Vizeli 2021); other CYP without in vivo effect.	moderate
MDMA	CYP2D6 (demethylation, that MDMA itself inhibits in ~2 h), CYP1A2, CYP2B6 and CYP2C19; COMT methylates HHMA.	CYP2D6 NM probable; CYP2C19 *1/*1; CYP2B6 *2/*4; COMT Val/Val (haplotype LPS/LPS, high activity); CYP1A2 *1F heterozygous.	Typical profile. Self-inhibition of CYP2D6 reduces the influence of the genotype after the first dose. In the studies (Vizeli 2017), the effects of genotypes were small. Repeated doses accumulate in a nonlinear way.	moderate
Ketamine/ sketamine	CYP2B6 (N-demethylation, main route) and CYP3A4 → norcetamine.	CYP2B6 *2/*4 (without *6/*18): normal to fast. CYP3A4 *1/*1; CYP3A5 *1/*3 (expressor). BDNF Val/Met.	Typical or slightly faster clearance. *6 (absent) is associated with slower clearance and more adverse effects. The ketamine bull has no pharmacogenetic recommendation.	moderate (CYP2B6*6) · weak (BDNF)
DMT / ayahuasca	DMT: MAO-A (so oral inactive without MAOIs); CYP2D6 minor. Harminal: CYP1A2 and CYP2D6; harmalin: CYP2D6. The β-carbolins are reversible IMAO-A.	CYP2D6 NM probable; CYP1A2 *1F A/C + rs2472297 C/C (normal to intermediate elimination); MAOA rs6323 T (marker; uVNTR not visible).	No strong change predicted. With the normal to intermediate CYP1A2, the harmonic should not be too prolonged. The main risk is the interactions of MAOIs: serotonergic, tyramine and lithium (see interactions).	weak
5-MeO-DMT	MAO-A (main); CYP2D6 → bufotenin (O-demethylation).	Probable CYP2D6 NM; marker MAOA.	Typical exposure. With harmalin (MAOIs + CYP2D6 inhibitor), exposure rises a lot; there are fatal cases described (Shen 2010).	moderate (interaction) · weak (genotype)
Ibogain	CYP2D6 → noribogain (long semicide); prolongs QT.	Likely CYP2D6 NM. Common loci of QT without risk alleles: NOS1AP rs12143842 C/C and rs10494366 T/T; KCNH2 K897T (rs1805123) T/G (1 allele associated with shorter QT); KCNQ1 rs2074238 C/C; SCN5A H558R (rs1805124) T/T.	Typical exposure to ibogain. Ibogain is associated with severe arrhythmias and deaths, even without genetic factors; QT SNPs do not exclude this risk.	strong (clinical cardiac risk) · weak (genotype)
Mescaline	Excreted largely unchanged; oxidative deamination (MAO and other amine-oxidases) → TMPAA.	No established pharmacogenetic gene.	No genetic prediction. Relevant interaction: lithium (classical psychedelic).	weak
THC (cannabis)	CYP2C9 → 11-OH-THC (main route); Secondary CYP3A4.	CYP2C9 *1/*3 (rs1057910 A/C): intermediate metabolizer. FAAH rs324420 C/A.	Exposure to THC probably higher than in *1/*1 (in *3/*3 was about 3 times higher). More anandamide (FAAH). It joins the "Psychedelic Effects of THC" score and the psychosis genes.	moderate (CYP2C9*3) · weak (FAAH)
CBD	CYP2C19 and CYP3A4 (metabolism); inhibits CYP3A4, CYP2C19 and CYP2C9.	CYP2C19 *1/*1; CYP3A4 *1/*1.	Typical metabolism. The most important is CBD as an inhibitor: it can rise sirolimus (rapamycin) and other substrates.	moderate (interaction)

Other extra SNPs (metabolism, receivers and QT)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
CYP1A2	rs762551	CA	chip	A	1	*1F (-163C>A), inducibility
CYP1A2/ CYP1B1	rs2472297	CC	chip	T	0	caffeine clearance
UGT1A9	rs72551330	TT	chip	C	0	*3 (M33T), reduced
UGT1A9	rs17868320	CC	chip	T	0	-2152C>T, higher expression
UGT1A9	rs6714486	TT	imputation DR2=1.00	A	0	-275T>A, higher expression
UGT1A1	rs887829	CT	chip	T	1	*28 marker (Gilbert)
MAOA	rs6323	T	chip	—	—	R297R (T941G; high activity marker)
MAOA	rs1137070	C	chip	—	—	D470D (marker)
MAOA	rs909525	T	chip	—	—	intronic (marker)
COMT	rs4680	GG	chip	A	0	Val158Met
COMT	rs4633	CC	chip	—	—	H62H (haplotype)
COMT	rs4818	GG	imputation DR2=1.00	—	—	L136L (haplotype LPS/APS/HPS)
COMT	rs6269	GG	chip	—	—	haplotype
HTR2A	rs6314	GA	chip	A	1	His452Tyr (in vitro response to psychedelics)
HTR2A	rs7997012	GG	chip	—	—	intronic (response to antidepressants)
HTR2A	rs6313	GG	chip	A	0	T102C
HTR2A	rs6311	CC	chip	T	0	-1438G>A
BDNF	rs6265	CT	chip	T	1	Val66Met
OPRM1	rs1799971	AG	chip	—	—	A118G
ANKK1/DRD2	rs1800497	GG	chip	—	—	Taq1A
DRD2	rs1076560	CC	chip	A	0	splicing D2S/D2L
FAAH	rs324420	CA	chip	A	1	P129T, lower activity
CNR1	rs1049353	CT	chip	T	1	1359G>A
OXTR	rs53576	GG	chip	—	—	intronic
OXTR	rs2254298	AA	chip	—	—	intronic
CLOCK	rs1801260	AG	chip	—	—	3111T>C
NOS1AP	rs12143842	CC	chip	T	0	Longer QT
NOS1AP	rs10494366	TT	chip	G	0	QT
KCNH2	rs1805123	TG	chip	G	1	K897T (shorter QT)
SCN5A	rs1805124	TT	chip	—	—	H558R
KCNQ1	rs2074238	CC	chip	T	0	Shorter QT

Interactions with the current stack

Crossing between the current stack (medicines, supplements and peptides) and the substances of the report. These interactions depend on the stack, not on genetics, and almost all of them weigh more than SNPs.

Stack	Substances	Interaction	Evidence	Priority
Lithium 5 mg/day (supplement)	LSD, psilocybin, DMT/ayahuasca, mescaline	In an online analysis of 62 reports of lithium with classic psychedelics, 47% described seizures and 39% needed medical care (Nayak 2021). The reports involved mainly therapeutic lithium; there are no data with 5 mg. In clinical trials, lithium is an exclusion criterion.	weak-moderate (reports)	high
Naltrexone in low dose 4.5 mg	Ketamine/sketamine	Naltrexone (50 mg) blocked the antidepressant effect of ketamine in a small trial (Williams 2018), but not in another one (Yoon 2019). No data with the low dose.	weak (contradictory results)	moderate
Retatrutide 8 mg/6 days	Oral substances (psilocybin, MDMA, LSD, ayahuasca)	GLP-1/GIP agonists delay gastric emptying: the onset of the effect may be slower and more unpredictable, which leads to redoing. Nausea and vomiting are added to those of ayahuasca and psilocybin.	mechanism (bulb)	moderate
Dapagliflozin 10 mg	MDMA; ayahuasca (vomiting); long sessions without eating	ISGLT2 increases volume loss and the risk of euglycemic ketoacidosis with dehydration, fasting, vomiting or acute disease (bulb alertness). MDMA causes hyperthermia and sweating.	mechanism (bulb)	moderate-high
Retatrutide + dapagliflozin + imeglumine + acarbose	Long sessions, fasting	With 4 drugs that lower glucose, eating little for many hours increases the risk of hypoglycemia (especially with retatrutide). With acarbose, hypoglycemia is treated with dextrose.	mechanism	moderate
Oxytocin 160 µg (peptide)	MDMA	MDMA releases oxytocin and vasopressin and is associated with hyponatremia, especially with a lot of water. Exogenous oxytocin has an antidiuretic effect. The added risk is plausible, but it is not studied.	mechanism	moderate
Tadalafil 5 mg (+ telmisartan 20 mg)	Alkyl nitrites (poppers); MDMA; ketamine	Tadalafil with nitrites: severe hypotension (absolute contraindication of the leaflet). MDMA and ketamine increase heart rate and cardiovascular load; with cardiovascular medication, the response can be unpredictable.	strong (nitrites) · mechanism (rest)	high (nitrites)
Rapamycin 4 mg/6 weeks	CBD; THC; ketamine	CBD inhibits CYP3A4: there are cases of high levels of tacrolimus and everolimus with CBD, and sirolimus follows the same pathway. Ketamine is a substrate of CYP3A4 without known relevant interaction with rapamycin.	moderate (CBD)	moderate
Isotretinoin 40 mg/week	All (mainly psychedelic and cannabis)	The bull of isotretinoin alerts to depression, psychosis and suicidal ideation (causality is debated). Psychedelics can destabilize mood. The genetic liver vulnerability (PNPLA3, Pi* <i>MZ</i>) and the hepatotoxicity of MDMA are added.	weak-moderate	moderate
Curcumin, quercetin, luteolin, apigenin (stack reported)	Psilocin, THC, ketamine, LSD	In vitro, they inhibit UGT1A9/UGT1A1, CYP2C9 and CYP3A4. They can increase the exposure to psilocin, THC (sound to CYP2C9*3) and CYP3A4 substrates. No clinical data with psychedelics.	weak (in vitro)	low
Melatonin (up to 100 mg)	Ketamine; sedatives	Sedation is added to the central nervous system depressants. With normal to intermediate CYP1A2, melatonin lasts the usual (0.3–1 mg).	weak	low
Without SSRIs/SNRIs, MAOIs, tryptans or tramadol in the stack	Ayahuasca, MDMA, 5-MeO-DMT	There are no drugs in the stack with more risk of serotonergic syndrome with MAOIs or MDMA. SSRIs also attenuate psychedelics.	—	(favourable)

Any change to the stack: suggestion (not applied), to decide with the doctor.

Mental Health Risk

In this section, and only in this section, the results are separated into favourable and unfavourable. Percentages above 75 = MORE LIKELY and below 25 = LESS LIKELY, as in the example report.

Unfavourable

- MORE LIKELY** **Bipolar Disorder:** More likely to have bipolar disease
- MORE LIKELY** **Borderline Personality:** More likely to have borderline disturbance
- MORE LIKELY** **Schizophrenia:** More likely to have schizophrenia

Favourable

None.

Typical / non-reproductive

- TYPICAL LIKELIHOOD** **THC and Psychosis:** Typical probability of psychotic symptoms with cannabis
- TYPICAL LIKELIHOOD** **Psychological Trauma:** Typical probability of PTSD
- NOT REPRODUCIBLE** **Non-Epileptic Seizures:** No published genetic data to reproduce this trait (proxy: neuroticism)
- TYPICAL LIKELIHOOD** **Anxiety:** Typical probability of anxiety
- TYPICAL LIKELIHOOD** **Panic Attacks:** Typical probability of panic attacks
- TYPICAL LIKELIHOOD** **Seizures:** Typical probability of seizures (epilepsy)

A percentile 88 means that the score is higher than that of 88% of the reference population. Even in the high percentiles, the relative risk increase is modest, and most people with these scores never develop the condition.

Bipolar Disorder

MORE LIKELY More likely to have bipolar disease

PGS002786: 936 730 of 948 996 variants with genotype (coverage 99%). the 79th percentile in the reference population (n=633).

High hereditary disease (up to 80%). In psychedelic trials, the personal or family history of bipolar disorder is a criterion for exclusion, because a single experience can precipitate mania. Score: PGS002786 (Gui 2022, SDPR, 949 thousand variants).

Top variants with risk allele (12)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
DUSP28	rs2953145	CC	chip	C	2	Risk allele in GWAS Catalog
DCTN5	rs420259	AG	chip	A	1	Risk allele in GWAS Catalog
TDRD9	rs11622475	CC	chip	C	2	Risk allele in GWAS Catalog
—	rs145410455	GG	imputation DR2=0.99	G	2	Risk allele in GWAS Catalog
SPPL3	rs58235352	GA	imputation DR2=1.00	G	1	Risk allele in GWAS Catalog
SORCS3	rs61867293	CT	imputation DR2=0.99	C	1	Risk allele in GWAS Catalog
MYH15	rs1531188	CC	imputation DR2=0.98	C	2	Risk allele in GWAS Catalog
PLXNB2	rs113872034	GG	imputation DR2=1.00	G	2	Risk allele in GWAS Catalog
EIF3M	rs143864773	TC	imputation DR2=0.98	T	1	Risk allele in GWAS Catalog
OLFM4	rs12552	AG	chip	A	1	Risk allele in GWAS Catalog
RARRES1	rs7430565	GA	chip	G	1	Risk allele in GWAS Catalog
EFNA5	rs55993664	CC	imputation DR2=0.98	C	2	Risk allele in GWAS Catalog

Top variants without risk allele (meaning opposite) (5)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
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CDC25B	rs3761218	TT	chip	C	0	Risk allele in GWAS Catalog
CCDC198	rs10134944	CC	chip	T	0	Risk allele in GWAS Catalog
SOX6	rs977509	TT	imputation DR2=1.00	C	0	Risk allele in GWAS Catalog
SOX5	rs4074723	AA	imputation DR2=1.00	C	0	Risk allele in GWAS Catalog
ENOX1	rs4143229	AA	imputation DR2=0.99	C	0	Risk allele in GWAS Catalog

Risk allele not published in the catalog (33)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
NRN1	rs4960155	CT	imputation DR2=0.96	—	—	Risk allele not published in the catalog
PSAPL1	rs7683874	GA	chip	—	—	Risk allele not published in the catalog
THRA	rs2314339	CC	chip	—	—	Risk allele not published in the catalog
HES6	rs2304672	GG	chip	—	—	Risk allele not published in the catalog
ARVCF	rs165599	GA	chip	—	—	Risk allele not published in the catalog
BCR	rs131690	GG	imputation DR2=1.00	—	—	Risk allele not published in the catalog
CMTM8	rs4276227	CC	imputation DR2=1.00	—	—	Risk allele not published in the catalog
BCR	rs131702	GG	imputation DR2=0.92	—	—	Risk allele not published in the catalog
WHRN	rs10982256	TT	chip	—	—	Risk allele not published in the catalog
SH2B3	rs3847953	CT	imputation DR2=0.92	—	—	Risk allele not published in the catalog
MAPK1	rs8136867	GA	chip	—	—	Risk allele not published in the catalog
TPH2	rs17110747	AA	imputation DR2=0.98	—	—	Risk allele not published in the catalog
THRA	rs2071427	CC	chip	—	—	Risk allele not published in the catalog
POU3F3	rs7570682	GA	chip	—	—	Risk allele not published in the catalog
HES6	rs2304669	TT	chip	—	—	Risk allele not published in the catalog
THRA	rs939348	CC	chip	—	—	Risk allele not published in the catalog
DPP10	rs1375144	AA	chip	—	—	Risk allele not published in the catalog
TPH2	rs1843809	GT	chip	—	—	Risk allele not published in the catalog
THRA	rs2269457	TT	chip	—	—	Risk allele not published in the catalog
PPARGC1B	rs7732671	GG	chip	—	—	Risk allele not published in the catalog
CHD9	rs1344484	TC	chip	—	—	Risk allele not published in the catalog
TRIB2	rs4027132	AA	chip	—	—	Risk allele not published in the catalog
SDCCAG8	rs6703335	GG	chip	—	—	Risk allele not published in the catalog
CHRNA7	rs6494223	TT	chip	—	—	Risk allele not published in the catalog
BICRAL	rs6458307	CC	chip	—	—	Risk allele not published in the catalog
NT5C2	rs11191580	TT	chip	—	—	Risk allele not published in the catalog
NFIA	rs7556462	CC	imputation DR2=1.00	—	—	Risk allele not published in the catalog
CLOCK	rs3805148	AC	imputation DR2=1.00	—	—	Risk allele not published in the catalog
CLOCK	rs4864542	CG	imputation DR2=1.00	—	—	Risk allele not published in the catalog
NFKB1	rs230529	CC	imputation DR2=1.00	—	—	Risk allele not published in the catalog
NPAS2	rs1562313	CC	chip	—	—	Risk allele not published in the catalog
MANBA	rs230535	CC	chip	—	—	Risk allele not published in the catalog
ZNF804A	rs1344706	CC	chip	—	—	Risk allele not published in the catalog

See "Reduction of risks" and "Recommendations in detail": suggestion (not applied).

THC and Psychosis

TYPICAL LIKELIHOOD Typical probability of psychotic symptoms with cannabis

Score of 6 of 6 variants (sum of effect alleles). In the reference population (n=633): the 70th percentile (mean position); 23% have exactly the same score.

Score of 6 variants associated with psychotic symptoms with cannabis (CNR1 rs1049353 T and rs12720071 T; FAAH rs2295633 A; COMT rs4680 G; DRD2 rs1076560 A; AKT1 rs2494732 C). Individual effects come from small studies; AKT1 C was associated with about 7 times more psychosis in daily smokers in a study, but replication is partial.

Effect allele (in the sense of result) (4)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
CNR1	rs1049353	CT	chip	T	1	less CB1 expression; more psychotic symptoms with cannabis
FAAH	rs2295633	AG	chip	A	1	G = more FAAH (protector); A : sense of risk
CNR1	rs12720071	TT	chip	T	2	psychotic symptoms with cannabis
COMT	rs4680	GG	chip	G	2	Val: plus COMT; risk of psychosis with cannabis

No effect allele (meaning contrary / reference) (2)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
TTC12	rs1076560	CC	chip	A	0	plus D2 signaling; psychosis with cannabis
AKT1	rs2494732	TT	chip	C	0	Psychosis in daily cannabis smokers

Borderline Personality

MORE LIKELY More likely to have borderline disturbance

Weighted score (ln OR of GWAS Catalog) with 20 of 20 top variants. the 77th percentile in the reference population (n=633). Low reliability: Small GWAS.

There is no published borderline disorder PGS. We used 28 Witt's GWAS associations 2017 ($p < 1 \times 10^{-5}$, no genome-wide), weighted by ln(OR). Some variants are rare, with very high OR and little precision. Low reliability.

Top variants with risk allele (12)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
DPYD	rs187785463	GG	imputation DR2=1.00	G	2	Risk allele in GWAS Catalog
MYCN	rs57726666	AG	imputation DR2=0.98	A	1	Risk allele in GWAS Catalog
SKP2	rs34416917	GG	chip	G	2	Risk allele in GWAS Catalog
SYT1	rs11110253	AA	imputation DR2=0.95	A	2	Risk allele in GWAS Catalog
FAM155A	rs9520569	CT	imputation DR2=0.82	T	1	Risk allele in GWAS Catalog
TRDN	rs2145737	TT	imputation DR2=0.97	T	2	Risk allele in GWAS Catalog
SNX9	rs6922614	TT	imputation DR2=1.00	T	2	Risk allele in GWAS Catalog
MAML3	rs13136239	GG	imputation DR2=0.98	G	2	Risk allele in GWAS Catalog
DPRX	rs8104156	GT	imputation DR2=0.93	T	1	Risk allele in GWAS Catalog
METTL21C	rs675828	AG	chip	G	1	Risk allele in GWAS Catalog
GDF6	rs11784341	CT	imputation DR2=0.82	T	1	Risk allele in GWAS Catalog
—	rs76695126	TC	imputation DR2=0.92	T	1	Risk allele in GWAS Catalog

Top variants without risk allele (meaning opposite) (8)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
GAP43	rs283386	AA	imputation DR2=0.95	G	0	Risk allele in GWAS Catalog
GAS1	rs7859734	TT	imputation DR2=0.96	C	0	Risk allele in GWAS Catalog
TWIST1	rs114497090	GG	imputation DR2=0.94	A	0	Risk allele in GWAS Catalog
—	rs115689122	TT	imputation DR2=0.87	A	0	Risk allele in GWAS Catalog
—	rs187058036	TT	imputation DR2=1.00	C	0	Risk allele in GWAS Catalog
FOXP1	rs6975373	CC	imputation DR2=0.89	T	0	Risk allele in GWAS Catalog
EME1	rs1985762	CC	imputation DR2=0.88	T	0	Risk allele in GWAS Catalog
TSPAN14	rs7074356	GG	imputation DR2=0.98	A	0	Risk allele in GWAS Catalog

SNP not found in the data (3)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
—	rs150592717	—	SNP not found in the data (Imputation DR2=0.00, below the threshold 0.8)	T	—	Risk allele in GWAS Catalog
—	rs62127626	—	SNP not found in the data (Imputation DR2=0.78, below the threshold 0.8)	A	—	Risk allele in GWAS Catalog
CLEC4C	rs113507694	—	SNP not found in the data (Imputation DR2=0.46, below the threshold 0.8)	G	—	Risk allele in GWAS Catalog

See "Reduction of risks" and "Recommendations in detail": suggestion (not applied).

Psychological Trauma

TYPICAL LIKELIHOOD Typical probability of PTSD

PGS005393: 18 401 of 53 705 variants with genotype (coverage 34%). the 59th percentile in the reference population (n=633).

Heritability up to 40%. Score: PGS005393 (Bugiga 2024, PRSice, 53 705 variants). Only 34% of the score variants were found in the data (chip + high-confidence imputation), which reduces reliability.

Top variants with risk allele (11)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
UST	rs10457838	CC	chip	C	2	Risk allele in GWAS Catalog
CSMD1	rs2616978	TT	chip	T	2	Risk allele in GWAS Catalog
KIAA1109	rs45510091	AA	imputation DR2=1.00	A	2	Risk allele in GWAS Catalog
TCF4	rs599550	AA	imputation DR2=0.99	A	2	Risk allele in GWAS Catalog
MLKL	rs62056018	AA	imputation DR2=0.89	A	2	Risk allele in GWAS Catalog
SOX6	rs931774	TT	imputation DR2=0.91	T	2	Risk allele in GWAS Catalog
SERPING1	rs2509805	TT	imputation DR2=0.98	T	2	Risk allele in GWAS Catalog
FAM120AOS	rs10992729	CT	imputation DR2=1.00	T	1	Risk allele in GWAS Catalog
—	rs11933210	TT	imputation DR2=0.96	T	2	Risk allele in GWAS Catalog
ZNF804A	rs62176173	GT	imputation DR2=0.98	T	1	Risk allele in GWAS Catalog
PTPN7	rs4511180	AA	chip	A	2	Risk allele in GWAS Catalog

Top variants without risk allele (meaning opposite) (7)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
—	rs58649573	CC	chip	T	0	Risk allele in GWAS Catalog
PTPN7	rs3100127	CC	chip	A	0	Risk allele in GWAS Catalog
POGK	rs2312236	CC	chip	T	0	Risk allele in GWAS Catalog
PRTFDC1	rs1033962	CC	chip	A	0	Risk allele in GWAS Catalog
TRIM27	rs145108206	GG	imputation DR2=1.00	A	0	Risk allele in GWAS Catalog
—	rs1246683	GG	imputation DR2=0.90	A	0	Risk allele in GWAS Catalog
MAPT	rs62056789	CC	imputation DR2=1.00	A	0	Risk allele in GWAS Catalog

Risk allele not published in the catalog (32)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
CRHR2	rs2267715	GG	chip	—	—	Risk allele not published in the catalog
RORA	rs8041061	GT	chip	—	—	Risk allele not published in the catalog
MAPT	rs12938031	GG	chip	—	—	Risk allele not published in the catalog
RGS2	rs4606	CG	chip	—	—	Risk allele not published in the catalog
ANKK1	rs1800497	GG	chip	—	—	Risk allele not published in the catalog
—	rs61793204	GG	chip	—	—	Risk allele not published in the catalog
TBC1D2	rs7866350	CC	chip	—	—	Risk allele not published in the catalog

FBLL1	rs10038727	GA	chip	—	—	Risk allele not published in the catalog
SH2D1B	rs386231	TT	chip	—	—	Risk allele not published in the catalog
NOS1AP	rs10918936	AG	chip	—	—	Risk allele not published in the catalog
TNF	rs1800629	GG	chip	—	—	Risk allele not published in the catalog
C1ORF226	rs12027674	AG	chip	—	—	Risk allele not published in the catalog
ARHGAP27	rs4792887	CC	chip	—	—	Risk allele not published in the catalog
BDNF	rs6265	CT	chip	—	—	Risk allele not published in the catalog
FAAH	rs324420	CA	chip	—	—	Risk allele not published in the catalog
NOS1	rs10744891	GG	chip	—	—	Risk allele not published in the catalog
ESR1	rs9340799	AA	chip	—	—	Risk allele not published in the catalog
IL1B	rs16944	AG	chip	—	—	Risk allele not published in the catalog
TERT	rs2736100	CC	chip	—	—	Risk allele not published in the catalog
ACE	rs4311	TC	chip	—	—	Risk allele not published in the catalog
IL6	rs1800795	GG	chip	—	—	Risk allele not published in the catalog
FAAH	rs2295633	AG	chip	—	—	Risk allele not published in the catalog
TULP1	rs3800373	AA	chip	—	—	Risk allele not published in the catalog
DUSP23	rs1130864	GG	chip	—	—	Risk allele not published in the catalog
ATP10B	rs17504106	GG	chip	—	—	Risk allele not published in the catalog
ADRB2	rs2400707	AG	chip	—	—	Risk allele not published in the catalog
SRR	rs4523957	TT	chip	—	—	Risk allele not published in the catalog
OXTR	rs53576	GG	chip	—	—	Risk allele not published in the catalog
UNC13C	rs73419609	AA	chip	—	—	Risk allele not published in the catalog
TTC12	rs2075652	GG	chip	—	—	Risk allele not published in the catalog
LRRC4C	rs10768747	GA	imputation DR2=0.99	—	—	Risk allele not published in the catalog
DUSP23	rs3091244	GG	chip	—	—	Risk allele not published in the catalog

Non-Epileptic Seizures

NOT REPRODUCIBLE No published genetic data to reproduce this trait (proxy: neuroticism)

PGS003565: 972 627 of 979 739 variants with genotype (coverage 99%). the 63rd percentile in the reference population (n=633).

The example report does not list variants for this trait, and there is no published GWAS or PGS of non-epileptic psychogenic seizures (the catalog associations for 'psychogenic disturbance' are almost fixed variants, without information). Unreproducible trait. As a context (proxy), we show PGS of neuroticism PGS003565 (Ding 2023), a trait associated with stress, anxiety and functional symptoms.

There are no GWAS published with enough variants to reproduce this trait. As an indicative proxy, the PGS of neuroticism PGS003565 was used: the 63rd percentile (TYPICAL). The variants of the example report are below, only with the genotype.

Schizophrenia

MORE LIKELY More likely to have schizophrenia

PGS002785: 950 324 of 964 422 variants with genotype (coverage 98%). the 88th percentile in the reference population (n=633).

Heritability of 70–80%. Score: PGS002785 (Gui 2022, SDPR, 964 thousand variants). This PGS only explains a fraction of the genetic risk; the use of psychoactives in adolescence, stress and family history tell a lot.

Top variants with risk allele (35)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
DPYD	rs1702294	CC	chip	C	2	Risk allele in GWAS Catalog
PILRB	rs41295924	CC	imputation DR2=0.89	C	2	Risk allele in GWAS Catalog

H4C13	rs140365013	GG	imputation DR2=1.00	G	2	Risk allele in GWAS Catalog
EFCAB6	rs76365544	GG	imputation DR2=0.98	G	2	Risk allele in GWAS Catalog
EDEM3	rs78444298	GG	chip	G	2	Risk allele in GWAS Catalog
ELAPOR2	rs137881681	GG	imputation DR2=0.96	G	2	Risk allele in GWAS Catalog
OR4C12	rs139161789	GG	imputation DR2=1.00	G	2	Risk allele in GWAS Catalog
DLC1	rs143092720	GG	imputation DR2=0.97	G	2	Risk allele in GWAS Catalog
RD3	rs117251211	GG	imputation DR2=1.00	G	2	Risk allele in GWAS Catalog
SIX3	rs79028395	GG	imputation DR2=0.81	G	2	Risk allele in GWAS Catalog
CCDC85A	rs74759822	TT	chip	T	2	Risk allele in GWAS Catalog
STAT6	rs61937595	CC	imputation DR2=0.99	C	2	Risk allele in GWAS Catalog
—	rs9456970	AA	imputation DR2=0.81	A	2	Risk allele in GWAS Catalog
TEF	rs143426938	GG	imputation DR2=0.95	G	2	Risk allele in GWAS Catalog
CCDC68	rs34751112	AA	chip	A	2	Risk allele in GWAS Catalog
LRRC46	rs11652374	TT	imputation DR2=1.00	T	2	Risk allele in GWAS Catalog
SFXN2	rs12571643	GG	imputation DR2=0.99	G	2	Risk allele in GWAS Catalog
CCDC68	rs12966547	GA	chip	G	1	Risk allele in GWAS Catalog
SP3	rs117073909	GG	imputation DR2=1.00	G	2	Risk allele in GWAS Catalog
SPATS2L	rs140001745	TT	imputation DR2=1.00	T	2	Risk allele in GWAS Catalog
RELN	rs62480723	CC	imputation DR2=0.96	C	2	Risk allele in GWAS Catalog
STAT6	rs34073963	GG	imputation DR2=0.87	G	2	Risk allele in GWAS Catalog
REX1BD	rs72999390	GG	imputation DR2=0.93	G	2	Risk allele in GWAS Catalog
ZNF365	rs11596514	TC	imputation DR2=0.93	T	1	Risk allele in GWAS Catalog
ARL5B	rs7893279	TT	chip	T	2	Risk allele in GWAS Catalog
TCF4	rs72934602	GG	imputation DR2=0.96	G	2	Risk allele in GWAS Catalog
GRM3	rs35274762	TT	imputation DR2=0.96	T	2	Risk allele in GWAS Catalog
ATP5MC3	rs62184532	GG	imputation DR2=1.00	G	2	Risk allele in GWAS Catalog
ACVR2A	rs114664644	GG	imputation DR2=1.00	G	2	Risk allele in GWAS Catalog
FUT9	rs117178087	CC	imputation DR2=0.98	C	2	Risk allele in GWAS Catalog
THAP8	rs3810450	TT	chip	T	2	Risk allele in GWAS Catalog
RELN	rs7341475	GG	chip	G	2	Risk allele in GWAS Catalog
TDRD3	rs139971826	GG	imputation DR2=0.83	G	2	Risk allele in GWAS Catalog
EPHX2	rs73229090	CC	chip	C	2	Risk allele in GWAS Catalog
ANKRD36	rs7575796	AA	imputation DR2=0.85	A	2	Risk allele in GWAS Catalog

Risk allele not published in the catalog (10)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
NRN1	rs4960155	CT	imputation DR2=0.96	—	—	Risk allele not published in the catalog
COMT	rs737865	GG	chip	—	—	Risk allele not published in the catalog
SH3YL1	rs6548238	CC	chip	—	—	Risk allele not published in the catalog
NAT16	rs1006507	TT	chip	—	—	Risk allele not published in the catalog
COMT	rs4680	GG	chip	—	—	Risk allele not published in the catalog
NRN1	rs9379002	TT	chip	—	—	Risk allele not published in the catalog
NRN1	rs3763180	TG	imputation DR2=0.90	—	—	Risk allele not published in the catalog
NRN1	rs1475157	AA	imputation DR2=0.99	—	—	Risk allele not published in the catalog
GRIN3A	rs149729514	GG	chip	—	—	Risk allele not published in the catalog
KLF9	rs11142387	CC	chip	—	—	Risk allele not published in the catalog

SNP not found in the data (5)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
CBLN3	rs146732081	—	SNP not found in the data (Imputation DR2=0.19, below the threshold 0.8)	C	—	Risk allele in GWAS Catalog
ARSA	rs113706174	—	SNP not found in the data (Imputation DR2=0.76, below the threshold 0.8)	C	—	Risk allele in GWAS Catalog
ZFP69	rs150863806	—	SNP not found in the data (Imputation DR2=0.77, below the threshold 0.8)	G	—	Risk allele in GWAS Catalog
RBFOX1	rs79478621	—	SNP not found in the data (Imputation DR2=0.79, below the threshold 0.8)	T	—	Risk allele in GWAS Catalog
FTCDNL1	rs76432012	—	SNP not found in the data (Imputation DR2=0.74, below the threshold 0.8)	T	—	Risk allele in GWAS Catalog

See "Reduction of risks" and "Recommendations in detail": suggestion (not applied).

Anxiety

TYPICAL LIKELIHOOD Typical anxiety probability

PGS018941: 706 532 of 729 735 variants with genotype (coverage 97%). the 26th percentile in the reference population (n=633).

Heritability of 30–65%. Main score: PGS018941 (generalized anxiety disorder, Chen 2025, 730 thousand variants). Complementary: PGS004451 (CID-10 F41, other anxiety disorders, which includes panic).

Top variants with risk allele (3)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
PREPL	rs1067327	CC	chip	C	2	Risk allele in GWAS Catalog
ZPLD1	rs1709393	CC	chip	C	2	Risk allele in GWAS Catalog
DMD	rs921896	C	chip	C	1	Risk allele in GWAS Catalog

Top variants without risk allele (meaning opposite) (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
NOX4	rs17221829	CC	imputation DR2=0.91	A	0	Risk allele in GWAS Catalog

Risk allele not published in the catalog (46)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
ATP8B4	rs2413998	AG	imputation DR2=1.00	—	—	Risk allele not published in the catalog
—	rs16838980	TT	imputation DR2=0.92	—	—	Risk allele not published in the catalog
DNAH8	rs4714177	AA	imputation DR2=0.85	—	—	Risk allele not published in the catalog
NUP107	rs11177321	GG	imputation DR2=0.94	—	—	Risk allele not published in the catalog
FKBP4	rs2302729	CC	chip	—	—	Risk allele not published in the catalog
RNF180	rs6295	CG	chip	—	—	Risk allele not published in the catalog
IL20RB	rs17374749	GG	imputation DR2=0.96	—	—	Risk allele not published in the catalog
PID1	rs10498237	AA	imputation DR2=1.00	—	—	Risk allele not published in the catalog
—	rs10092548	AG	imputation DR2=1.00	—	—	Risk allele not published in the catalog
C6ORF118	rs9295300	AG	imputation DR2=1.00	—	—	Risk allele not published in the catalog
NOX4	rs10830352	GG	imputation DR2=0.96	—	—	Risk allele not published in the catalog
GABRG2	rs211037	CC	chip	—	—	Risk allele not published in the catalog
MARCHF4	rs955816	GG	chip	—	—	Risk allele not published in the catalog
IRX6	rs2397376	CC	imputation DR2=1.00	—	—	Risk allele not published in the catalog
HTR2A	rs12584920	GG	chip	—	—	Risk allele not published in the catalog
COMT	rs4680	GG	chip	—	—	Risk allele not published in the catalog
ERCC6L2	rs7867155	CT	imputation DR2=0.94	—	—	Risk allele not published in the catalog
COMT	rs4633	CC	chip	—	—	Risk allele not published in the catalog

SLC6A2	rs3785151	GG	imputation DR2=0.85	—	—	Risk allele not published in the catalog
CES1	rs1566652	GT	imputation DR2=1.00	—	—	Risk allele not published in the catalog
GAD1	rs3828275	CC	chip	—	—	Risk allele not published in the catalog
GAD1	rs701492	TT	chip	—	—	Risk allele not published in the catalog
GAD1	rs769407	CC	chip	—	—	Risk allele not published in the catalog
GAD1	rs3791878	TT	imputation DR2=1.00	—	—	Risk allele not published in the catalog
IL18R1	rs2058622	GG	chip	—	—	Risk allele not published in the catalog
GAD1	rs3791851	CC	chip	—	—	Risk allele not published in the catalog
CAMTA1	rs11120917	CC	chip	—	—	Risk allele not published in the catalog
OR5P3	rs7112002	CC	imputation DR2=0.95	—	—	Risk allele not published in the catalog
SRBD1	rs2344662	AA	chip	—	—	Risk allele not published in the catalog
ADRB1	rs1034258	AA	imputation DR2=0.83	—	—	Risk allele not published in the catalog
SSH2	rs6354	GT	chip	—	—	Risk allele not published in the catalog
ESR1	rs9340799	AA	chip	—	—	Risk allele not published in the catalog
ESR1	rs2234693	TT	chip	—	—	Risk allele not published in the catalog
AKAP6	rs17406568	GG	imputation DR2=0.84	—	—	Risk allele not published in the catalog
OSCP1	rs906228	AA	imputation DR2=0.99	—	—	Risk allele not published in the catalog
AGPAT4	rs3798943	TC	imputation DR2=0.86	—	—	Risk allele not published in the catalog
CCNY	rs2086153	CT	imputation DR2=0.93	—	—	Risk allele not published in the catalog
COX7B2	rs6447514	TG	chip	—	—	Risk allele not published in the catalog
DDT	rs755622	CC	imputation DR2=0.98	—	—	Risk allele not published in the catalog
TULP1	rs3800373	AA	chip	—	—	Risk allele not published in the catalog
RGS2	rs10801153	GA	chip	—	—	Risk allele not published in the catalog
RNF220	rs12138940	GG	imputation DR2=0.96	—	—	Risk allele not published in the catalog
MC4R	rs10871777	AA	chip	—	—	Risk allele not published in the catalog
TBL1X	rs5934574	T	chip	—	—	Risk allele not published in the catalog
TACR1	rs3771841	GG	chip	—	—	Risk allele not published in the catalog
DSCAM	rs1040315	AA	imputation DR2=0.99	—	—	Risk allele not published in the catalog

Panic Attacks

TYPICAL LIKELIHOOD Typical probability of panic attacks

PGS004451: 1 036 079 of 1 059 939 variants with genotype (coverage 98%). the 55th percentile in the reference population (n=633).

There is no PGS of isolated panic disorder. Main score: PGS004451 (CID-10 F41: panic disorder, generalized and mixed anxiety; Jung 2024, LDpred2, 1.06 million variants). Complementary: 5 loci of the GWAS panic (Forstner 2021), weighted by ln(OR).

Top variants with risk allele (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
LGSN	rs6914428	GA	imputation DR2=0.99	G	1	Risk allele in GWAS Catalog

Top variants without risk allele (meaning opposite) (4)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
SNURF	rs2554444	AA	imputation DR2=0.88	T	0	Risk allele in GWAS Catalog
TFAP2C	rs79919349	GG	chip	A	0	Risk allele in GWAS Catalog
SMAD1	rs144783209	GG	imputation DR2=0.98	T	0	Risk allele in GWAS Catalog
SUSD1	rs41280169	CC	chip	T	0	Risk allele in GWAS Catalog

Risk allele not published in the catalog (9)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
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TMEM132D	rs11060369	AC	imputation DR2=0.98	—	—	Risk allele not published in the catalog
BLMH	rs140701	CC	chip	—	—	Risk allele not published in the catalog
BLMH	rs4583306	AA	imputation DR2=1.00	—	—	Risk allele not published in the catalog
SSH2	rs6354	GT	chip	—	—	Risk allele not published in the catalog
TPH1	rs1800532	GT	chip	—	—	Risk allele not published in the catalog
COMT	rs4680	GG	chip	—	—	Risk allele not published in the catalog
HTR2A	rs6313	GG	chip	—	—	Risk allele not published in the catalog
CCK	rs1799923	GG	imputation DR2=0.99	—	—	Risk allele not published in the catalog
CERS5	rs685012	TT	imputation DR2=0.97	—	—	Risk allele not published in the catalog

SNP not found in the data (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
—	rs25531	—	SNP not found in the data	—	—	Risk allele not published in the catalog

Seizures

TYPICAL LIKELIHOOD Typical probability of seizures (epilepsy)

PGS004881: 599 669 of 605 432 variants with genotype (coverage 99%). the 70th percentile in the reference population (n=633).

Some psychedelics and MDMA (high doses) lower the convulsive threshold; lithium with classical psychedelics was associated with seizures. Score: PGS004881 (epilepsy, Jermy 2024, 605 thousand variants).

Top variants with risk allele (15)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
ATXN1	rs68082256	GA	imputation DR2=0.98	A	1	Risk allele in GWAS Catalog
CDK5RAP3	rs4794333	TC	chip	C	1	Risk allele in GWAS Catalog
KRTAP8-1	rs2833098	GA	imputation DR2=0.99	G	1	Risk allele in GWAS Catalog
GABRA2	rs11943905	CT	imputation DR2=0.98	T	1	Risk allele in GWAS Catalog
ZEB2	rs10496964	CC	chip	C	2	Risk allele in GWAS Catalog
COPZ2	rs72823592	GA	chip	G	1	Risk allele in GWAS Catalog
CSMD2	rs771390	CC	imputation DR2=0.97	C	2	Risk allele in GWAS Catalog
TTC21B	rs11890028	TG	chip	T	1	Risk allele in GWAS Catalog
GOLIM4	rs111577701	CC	imputation DR2=0.93	C	2	Risk allele in GWAS Catalog
TTC21B	rs6732655	TT	chip	T	2	Risk allele in GWAS Catalog
VRK2	rs13026414	CT	chip	C	1	Risk allele in GWAS Catalog
SCN1A	rs12987787	TC	chip	T	1	Risk allele in GWAS Catalog
VRK2	rs1402398	GA	imputation DR2=0.99	G	1	Risk allele in GWAS Catalog
THEMIS	rs13200150	AG	imputation DR2=1.00	G	1	Risk allele in GWAS Catalog
NEMP2	rs887696	CC	imputation DR2=0.97	C	2	Risk allele in GWAS Catalog

Top variants without risk allele (meaning opposite) (13)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
PCDH7	rs1044352	GG	imputation DR2=1.00	T	0	Risk allele in GWAS Catalog
TRIM36	rs4596374	CC	chip	T	0	Risk allele in GWAS Catalog
FMN2	rs12059546	AA	chip	G	0	Risk allele in GWAS Catalog
PTGS2	rs12720541	GG	imputation DR2=1.00	T	0	Risk allele in GWAS Catalog
MMP8	rs1939012	CC	chip	T	0	Risk allele in GWAS Catalog
PCDH7	rs28498976	GG	chip	A	0	Risk allele in GWAS Catalog
GABRA2	rs535066	TT	chip	G	0	Risk allele in GWAS Catalog
UBXN2A	rs4665630	TT	chip	C	0	Risk allele in GWAS Catalog

IQCM	rs10030601	TT	imputation DR2=1.00	C	0	Risk allele in GWAS Catalog
VRK2	rs2717068	AC	chip	T	0	Risk allele in GWAS Catalog
MAST4	rs39861	AG	chip	C	0	Risk allele in GWAS Catalog
VRK2	rs2947349	AA	chip	C	0	Risk allele in GWAS Catalog
TRIM36	rs55670112	AA	chip	C	0	Risk allele in GWAS Catalog

Mental Health Genes

Gene activity: no good/bad judgment.

DISC1 (Mental Health)

TYPICAL ACTIVITY Predisposition for typical DISC1 activity

Score of 10 of 10 variants (sum of effect alleles). In the reference population (n=633): the 15th percentile (mean position); 22% have exactly the same score.

DISC1 coordinates the development and binding of neurons. We counted the 10 alleles associated with schizophrenia in the example report (rs821616 T, rs821617 A, rs821633 T, rs1538979 T, rs821597 A, rs6675281 T, rs2255340 T, rs2738864 T, rs2509382 C, rs3737597 A). More alleles mean, in the model of the example, less activity of DISC1.

Effect allele (in the sense of result) (4)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
DISC1	rs821597	GA	imputation DR2=0.99	A	1	risk
DISC1	rs821616	AT	imputation DR2=1.00	T	1	risk of schizophrenia
DISC1	rs821617	AG	imputation DR2=1.00	A	1	risk
DISC1	rs821633	TC	chip	T	1	risk

No effect allele (meaning contrary / reference) (6)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
DISC1	rs1538979	CC	imputation DR2=0.93	T	0	risk
DISC1	rs2255340	CC	imputation DR2=0.90	T	0	risk
DISC1	rs2738864	CC	imputation DR2=0.90	T	0	risk
—	rs2509382	GG	imputation DR2=0.95	C	0	risk
DISC1	rs3737597	GG	chip	A	0	risk
DISC1	rs6675281	CC	imputation DR2=0.99	T	0	risk

NRG1 (Mental Health)

LOWER ACTIVITY Predisposition for lower NRG1 activity

Score of 6 of 6 variants (sum of effect alleles). In the reference population (n=633): the 86th percentile (mean position); 12% have exactly the same score.

The neuregulin 1 participates in the formation of inhibitory synapses. We counted the 7 alleles of the example report (rs62510682 G, rs10503929 T, rs2954041 T, rs35753505 C, rs17603876 A, rs6994992 T, rs3924999 G). More alleles mean, in the model of the example, less activity of NRG1.

Effect allele (in the sense of result) (5)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
NRG1	rs10503929	TC	imputation DR2=0.96	T	1	risk
NRG1	rs3924999	GA	chip	G	1	risk
NRG1	rs17603876	AA	imputation DR2=0.95	A	2	risk
NRG1	rs6994992	TT	chip	T	2	risk
NRG1	rs35753505	CC	chip	C	2	risk

No effect allele (meaning contrary / reference) (1)

Gene	SNP	Genotype	Source	Effect allele	Copies	Effect/note
NRG1	rs2954041	GG	imputation DR2=1.00	T	0	risk

SNP not found in the data (1)

Gene	SNP	Genotype	Source	Copies
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				Effect allele		Effect/ note
NRG1	rs62510682	—	SNP not found in the data (Imputation DR2=0.00, below the threshold 0.8)	G	—	risk

Recommendations in detail

All entries: suggestion (not applied).

1. Psychotherapy (Psychotherapy; dose of the example: 1 hour)

Targeted risk	Sérgio F result	Impact	Evidence
Bipolar Disorder	MORE LIKELY	5/5	5/5
Psychological Trauma	TYPICAL LIKELIHOOD	5/5	5/5

Context (compatible with the stack): No known relevant interaction with the current stack. Suggestion (not applied).

2. Cognitive behavioral therapy (TCC) (Cognitive-Behavioral Therapy (CBT); example dose: —)

Targeted risk	Sérgio F result	Impact	Evidence
Bipolar Disorder	MORE LIKELY	3/5	3/5
Borderline Personality	MORE LIKELY	0/5	0/5
Psychological Trauma	TYPICAL LIKELIHOOD	5/5	5/5

Context (compatible with the stack): No known relevant interaction with the current stack. Suggestion (not applied).

3. Omega-3 (fish oil) (Omega-3 (Fish Oil); dose of the example: 500 mg)

Targeted risk	Sérgio F result	Impact	Evidence
Bipolar Disorder	MORE LIKELY	2/5	3/5
BDNF	SLIGHTLY LOWER LEVELS	2/5	2/5
Borderline Personality	MORE LIKELY	2/5	2/5
Psychological Trauma	TYPICAL LIKELIHOOD	2/5	2/5

Background (already in the stack): It is already part of the stack (EPA 1125 + DHA 875 mg), above the 500 mg of the example. The dose is managed in the plan of supplements; nothing to change here. Suggestion (not applied).

4. Solar exposure (Sunlight Exposure; dose of the example: 20 minutes)

Targeted risk	Sérgio F result	Impact	Evidence
Bipolar Disorder	MORE LIKELY	4/5	3/5

Context (carefulness or context): Nevi in the 88th percentile and isotretinoin (photosensitivity): prefer morning light, short exposures and no burn. Suggestion (not applied).

5. Vitamin C (Vitamin C; dose of the example: 500 mg)

Targeted risk	Sérgio F result	Impact	Evidence
Bipolar Disorder	MORE LIKELY	3/5	3/5

Background (already in the stack): In the stack (500 mg liposomal, between meals due to HFE H63D). Suggestion (not applied).

6. Yoga (Yoga; dose of the example: 30 minutes)

Targeted risk	Sérgio F result	Impact	Evidence
Bipolar Disorder	MORE LIKELY	1/5	1/5
BDNF	SLIGHTLY LOWER LEVELS	2/5	2/5
Borderline Personality	MORE LIKELY	0/5	0/5
Psychological Trauma	TYPICAL LIKELIHOOD	3/5	3/5

Context (compatible with the stack): No known relevant interaction with the current stack. Suggestion (not applied).

7. Dance (Dance; dose of the example: 30 minutes)

Targeted risk	Sérgio F result	Impact	Evidence
Borderline Personality	MORE LIKELY	2/5	2/5
Psychological Trauma	TYPICAL LIKELIHOOD	3/5	3/5

Context (compatible with the stack): No known relevant interaction with the current stack. *Suggestion (not applied).*

8. DHA (Omega-3) (DHA (Omega-3); dose of the example: 200 mg)

Targeted risk	Sérgio F result	Impact	Evidence
BDNF	SLIGHTLY LOWER LEVELS	2/5	2/5
Psychological Trauma	TYPICAL LIKELIHOOD	2/5	2/5

Background (already in the stack): Already included in the fish oil of the stack. *Suggestion (not applied).*

9. Modification of attentional bias (Attention Bias Modification; dose of the example: —)

Targeted risk	Sérgio F result	Impact	Evidence
Psychological Trauma	TYPICAL LIKELIHOOD	4/5	3/5

Context (compatible with the stack): No known relevant interaction with the current stack. *Suggestion (not applied).*

10. Mindfulness based cognitive therapy (MBCT) (Mindfulness-Based Cognitive Therapy (MBCT); example dose: 2 hours)

Targeted risk	Sérgio F result	Impact	Evidence
Psychological Trauma	TYPICAL LIKELIHOOD	4/5	3/5

Context (compatible with the stack): No known relevant interaction with the current stack. *Suggestion (not applied).*

11. Meditation (Meditation; dose of the example: 30 minutes)

Targeted risk	Sérgio F result	Impact	Evidence
Bipolar Disorder	MORE LIKELY	1/5	1/5
Borderline Personality	MORE LIKELY	0/5	0/5
Psychological Trauma	TYPICAL LIKELIHOOD	3/5	3/5

Context (compatible with the stack): No known relevant interaction with the current stack. *Suggestion (not applied).*

12. Mindfulness (Mindfulness; dose of example: 30 minutes)

Targeted risk	Sérgio F result	Impact	Evidence
Borderline Personality	MORE LIKELY	0/5	0/5
Non-Epileptic Seizures	NOT REPRODUCIBLE	1/5	1/5
Psychological Trauma	TYPICAL LIKELIHOOD	3/5	3/5

Context (compatible with the stack): No known relevant interaction with the current stack. *Suggestion (not applied).*

13. Mindfulness-based stress reduction (MBSR) (Mindfulness-Based Stress Reduction (MBSR); example dose: 2 hours)

Targeted risk	Sérgio F result	Impact	Evidence
Borderline Personality	MORE LIKELY	0/5	0/5
Psychological Trauma	TYPICAL LIKELIHOOD	3/5	3/5

Context (compatible with the stack): No known relevant interaction with the current stack. *Suggestion (not applied).*

14. Biofeedback (Biofeedback; dose of the example: —)

Targeted risk	Sérgio F result	Impact	Evidence
Psychological Trauma	TYPICAL LIKELIHOOD	3/5	3/5

Context (compatible with the stack): No known relevant interaction with the current stack. *Suggestion (not applied).*

15. Strength training (Strength Training; dose of the example: 48 hours)

Targeted risk	Sérgio F result	Impact	Evidence
Psychological Trauma	TYPICAL LIKELIHOOD	3/5	3/5

Background (already in the stack): Already indicated in the genetic plane (fragility in the 82nd percentile). *Suggestion (not applied).*

16. Art therapy (Art Therapy; dose of the example: 1 hour)

Targeted risk	Sérgio F result	Impact	Evidence
Bipolar Disorder	MORE LIKELY	1/5	1/5
Borderline Personality	MORE LIKELY	1/5	1/5
Psychological Trauma	TYPICAL LIKELIHOOD	2/5	1/5

Context (compatible with the stack): No known relevant interaction with the current stack. *Suggestion (not applied).*

17. Animal assisted therapy (Pet Therapy; dose of the example: 30 minutes)

Targeted risk	Sérgio F result	Impact	Evidence
Psychological Trauma	TYPICAL LIKELIHOOD	3/5	2/5

Context (compatible with the stack): No known relevant interaction with the current stack. *Suggestion (not applied).*

18. EFT (tapping) (EFT Tapping; dose of the example: 20 minutes)

Targeted risk	Sérgio F result	Impact	Evidence
Psychological Trauma	TYPICAL LIKELIHOOD	3/5	2/5

Context (compatible with the stack): No known relevant interaction with the current stack. *Suggestion (not applied).*

19. Transcendental meditation (Transcendental Meditation; dose of the example: 20 minutes)

Targeted risk	Sérgio F result	Impact	Evidence
Psychological Trauma	TYPICAL LIKELIHOOD	3/5	2/5

Context (compatible with the stack): No known relevant interaction with the current stack. *Suggestion (not applied).*

20. Relaxation applied (Applied Relaxation; dose of the example: 30 minutes)

Targeted risk	Sérgio F result	Impact	Evidence
Psychological Trauma	TYPICAL LIKELIHOOD	2/5	2/5

Context (compatible with the stack): No known relevant interaction with the current stack. *Suggestion (not applied).*

21. Acceptance therapy and commitment (ACT) (Acceptance and Commitment Therapy (ACT); example dose: —)

Targeted risk	Sérgio F result	Impact	Evidence
Bipolar Disorder	MORE LIKELY	1/5	1/5
Borderline Personality	MORE LIKELY	0/5	0/5
Psychological Trauma	TYPICAL LIKELIHOOD	1/5	1/5

Context (compatible with the stack): No known relevant interaction with the current stack. *Suggestion (not applied).*

22. Omega-3 food (Dietary Omega-3 Fatty Acids; dose of the example: —)

Targeted risk	Sérgio F result	Impact	Evidence
Bipolar Disorder	MORE LIKELY	1/5	1/5
Borderline Personality	MORE LIKELY	0/5	0/5

Background (already in the stack): The genetic plan already includes small fish 3x/week. *Suggestion (not applied).*

23. Black chocolate (Dark Chocolate; dose of example: —)

Targeted risk	Sérgio F result	Impact	Evidence
Bipolar Disorder	MORE LIKELY	1/5	1/5
BDNF	SLIGHTLY LOWER LEVELS	0/5	0/5

Context (carefulness or context): It has caffeine and theobromine; by sleep (insomnia p89), only in the morning and in little quantity. *Suggestion (not applied).*

24. Lithium orate (Lithium Orote; dose of the example: 5 mg)

Targeted risk	Sérgio F result	Impact	Evidence
Bipolar Disorder	MORE LIKELY	1/5	1/5

Background (already in the stack): It already takes lithium 5 mg. See interactions: lithium with classical psychedelics is associated with seizures. Discuss with the doctor before any psychedelic. *Suggestion (not applied).*

25. Dry fruits (Nuts; dose of example: 3 oz)

Targeted risk	Sérgio F result	Impact	Evidence
Bipolar Disorder	MORE LIKELY	1/5	1/5

Context (compatible with the stack): No known relevant interaction with the current stack. *Suggestion (not applied).*

26. Whole grains (Whole Grains; dose of the example: —)

Targeted risk	Sérgio F result	Impact	Evidence
Bipolar Disorder	MORE LIKELY	1/5	1/5

Context (compatible with the stack): No known relevant interaction with the current stack. *Suggestion (not applied).*

27. Scaled (Climbing; dose of the example: 1 hour)

Targeted risk	Sérgio F result	Impact	Evidence
Psychological Trauma	TYPICAL LIKELIHOOD	1/5	2/5

Context (compatible with the stack): No known relevant interaction with the current stack. *Suggestion (not applied).*

28. Tryptophan (Tryptophan; dose of the example: 500 mg)

Targeted risk	Sérgio F result	Impact	Evidence
Bipolar Disorder	MORE LIKELY	2/5	1/5

Background (not recommended in this context): Do not combine with MAOIs (ayahuasca, harmala), MDMA or other serotonergic (serotonergic syndrome). No genetic basis to add it. *Suggestion (not applied).*

29. Music therapy (Music Therapy; dose of example: 30 minutes)

Targeted risk	Sérgio F result	Impact	Evidence
Psychological Trauma	TYPICAL LIKELIHOOD	1/5	1/5

Context (compatible with the stack): No known relevant interaction with the current stack. *Suggestion (not applied).*

30. Lavender essential oil (Lavender Essential Oil; dose of the example: 2 drops)

Targeted risk	Sérgio F result	Impact	Evidence
Psychological Trauma	TYPICAL LIKELIHOOD	1/5	1/5

Context (compatible with the stack): No known relevant interaction with the current stack. *Suggestion (not applied).*

31. Diaphragmatic breathing (Diaphragmatic Breathing; dose of the example: 10 minutes)

Targeted risk	Sérgio F result	Impact	Evidence
Psychological Trauma	TYPICAL LIKELIHOOD	1/5	1/5

Context (compatible with the stack): No known relevant interaction with the current stack. *Suggestion (not applied).*

32. Kundalini yoga (Kundalini Yoga; dose of the example: 1 hour)

Targeted risk	Sérgio F result	Impact	Evidence
Psychological Trauma	TYPICAL LIKELIHOOD	1/5	1/5

Context (compatible with the stack): No known relevant interaction with the current stack. *Suggestion (not applied).*

33. Intensive light therapy in the morning (Morning Bright Light Therapy; dose of the example: 20 minutes)

Targeted risk	Sérgio F result	Impact	Evidence
Psychological Trauma	TYPICAL LIKELIHOOD	1/5	1/5

Context (carefulness or context): Useful for the circadian rhythm. The intense light only triggers mania in those who have bipolar disease; bipolar PGS (p79) is not a diagnosis. *Suggestion (not applied).*

34. N-back training (N-Back Training; dose of example: 20 minutes)

Targeted risk	Sérgio F result	Impact	Evidence
Psychological Trauma	TYPICAL LIKELIHOOD	1/5	1/5

Context (compatible with the stack): No known relevant interaction with the current stack. *Suggestion (not applied).*

35. Mindfulness meditation (Mindfulness Meditation; dose of example: 30 minutes)

Targeted risk	Sérgio F result	Impact	Evidence
BDNF	SLIGHTLY LOWER LEVELS	0/5	0/5

Context (compatible with the stack): No known relevant interaction with the current stack. *Suggestion (not applied).*

36. Ginkgo (Ginkgo; dose of the example: 120 mg)

Targeted risk	Sérgio F result	Impact	Evidence
BDNF	SLIGHTLY LOWER LEVELS	0/5	0/5

Context (carefulness or context): Variable effect in CYP2C19/CYP3A4 (rapamycin) and hemorrhagic risk added to the Omega-3; evidence 0/5 for BDNF. *Suggestion (not applied).*

37. Curcumin (Curcumin; dose of the example: 500 mg)

Targeted risk	Sérgio F result	Impact	Evidence
BDNF	SLIGHTLY LOWER LEVELS	0/5	0/5

Background (not recommended in this context): Pause is a suggestion (not applied; the real stack has not changed) of the genetic plan (liver: PNPLA3, Pi*MZ; UGT1A1*28; interaction with rapamycin). Do not resume for this reason. *Suggestion (not applied).*

38. D-manose (D-Mannose; dose of the example: 2g)

Targeted risk	Sérgio F result	Impact	Evidence
Bipolar Disorder	MORE LIKELY	0/5	0/5

Background (not recommended in this context): Evidence 0/5 (mendelian randomization only); without indication. *Suggestion (not applied).*

39. Bacopa (Bacopa; dose of the example: 300 mg)

Targeted risk	Sérgio F result	Impact	Evidence
BDNF	SLIGHTLY LOWER LEVELS	0/5	0/5

Context (carefulness or context): 0/5 evidence for BDNF; with the genetic liver, avoid new extracts without need. *Suggestion (not applied).*

40. Cocoa (Cocoa; dose of the example: —)

Targeted risk	Sérgio F result	Impact	Evidence
BDNF	SLIGHTLY LOWER LEVELS	0/5	0/5

Context (carefulness or context): As black chocolate: caffeine/theobromine, in the morning. *Suggestion (not applied).*

41. Green tea (extract) (Green Tea; dose of the example: 400 mg)

Targeted risk	Sérgio F result	Impact	Evidence
BDNF	SLIGHTLY LOWER LEVELS	0/5	0/5

Background (not recommended in this context): The genetic plan advises EGCG extracts (liver; catechol with COMT Val/Val). Tea as a drink in the morning does not have this problem. *Suggestion (not applied).*

42. Time in nature (Spend Time in Nature; dose of the example: 2 hours)

Targeted risk	Sérgio F result	Impact	Evidence
BDNF	SLIGHTLY LOWER LEVELS	0/5	0/5

Context (compatible with the stack): No known relevant interaction with the current stack. *Suggestion (not applied).*

43. Resilient starch (Resistant Starch; dose of the example: 40 g Hi-Maize)

Targeted risk	Sérgio F result	Impact	Evidence
BDNF	SLIGHTLY LOWER LEVELS	0/5	0/5

Context (carefulness or context): With acarbose, more fermentation and gases; the plane already rises the fiber (psyllium and PHGG). *Suggestion (not applied).*

44. Progressive muscle relaxation (Progressive Muscle Relaxation; dose of example: 10 minutes)

Targeted risk	Sérgio F result	Impact	Evidence
No DNA risks associated in the example report.			

Context (compatible with the stack): No known relevant interaction with the current stack. *Suggestion (not applied).*

45. Food zinc (Dietary Zinc; dose of the example: —)

Targeted risk	Sérgio F result	Impact	Evidence
No DNA risks associated in the example report.			

Background (already in the stack): The genetic plan already predicts total zinc of 20–25 mg/day with copper. *Suggestion (not applied).*

46. Galphimia (Galphimia; dose of the example: 350 mg)

Targeted risk	Sérgio F result	Impact	Evidence
No DNA risks associated in the example report.			

Context (carefulness or context): Few security and interaction data; low priority. *Suggestion (not applied).*

47. Nidra (Yoga Nidra; dose of the example: 30 minutes)

Targeted risk	Sérgio F result	Impact	Evidence
Psychological Trauma	TYPICAL LIKELIHOOD	0/5	0/5

Context (compatible with the stack): No known relevant interaction with the current stack. *Suggestion (not applied).*

48. CBD (CBD; dose of the example: 10 mg)

Targeted risk	Sérgio F result	Impact	Evidence
No DNA risks associated in the example report.			

Background (not recommended in this context): CBD inhibits CYP3A4, CYP2C19 and CYP2C9: it can rise sirolimus (rapamycin); with CYP2C9 *1/*3, also more THC if there is a mixture. Do not add without medical follow-up. *Suggestion (not applied).*

49. Rhyke therapy (Laughter Therapy; dose of example: 30 minutes)

Targeted risk	Sérgio F result	Impact	Evidence
No DNA risks associated in the example report.			

Context (compatible with the stack): No known relevant interaction with the current stack. *Suggestion (not applied).*

50. Massage (Massage; dose of the example: 30 minutes)

Targeted risk	Sérgio F result	Impact	Evidence
No DNA risks associated in the example report.			

Context (compatible with the stack): No known relevant interaction with the current stack. *Suggestion (not applied).*

Risk reduction

Recommendations of this report, based on genetic results and stack. Clinical framework and risk reduction; they do not encourage use.

1. Use psychedelics only in clinical context or authorized trial

The legal status depends on the country and most of these substances are controlled. Esketamine is approved for resistant depression; psilocybin and MDMA are in trials. The trials make psychiatric screening and exclude those who have a personal or family history of psychosis or bipolar disease. The PGS of schizophrenia (p88) and bipolar (p79) are not criteria for exclusion or diagnosis, but they weigh in favor of prudence.

Evidence: strong (clinical practice) • *suggestion (not applied)*

2. Talk to the doctor about lithium before any classic psychedelic

The described interaction (convulsions) makes lithium a contraindication in the research protocols. There are no data with 5 mg/day. The suspension or not is a medical decision.

Evidence: moderate • *suggestion (not applied)*

3. Clinical pharmacogenetic test with CNV of CYP2D6

The chip does not see deletions (*5) or duplications. This counts for LSD, MDMA, 5-MeO-DMT, ibogain and harmalin.

Evidence: moderate • *suggestion (not applied)*

4. Do not use ibogain

Risk of severe arrhythmia and sudden death, even with normal ECG. Common risk-free QT SNPs do not exclude danger.

Evidence: strong • *suggestion (not applied)*

5. Never combine tadalafil with alkyl nitrites ("poppers")

Absolute contraindication of the bull: severe hypotension.

Evidence: strong • *suggestion (not applied)*

6. Hydration, electrolytes and "rule of days of disease" with dapagliflozin and retatrutide

In prolonged fasting, vomiting or hyperthermia (MDMA, ayahuasca), the risk of dehydration and euglycemic ketoacidosis rises. The package leaflet provides suspension of iSGLT2 in these situations; the decision is medical.

Evidence: moderate (bully) • *suggestion (not applied)*

7. Avoid THC and high power cannabis

CYP2C9 *1/*3 rises the exposure to THC. The score of psychedelic effects of THC is "Increased" and the PGS of schizophrenia in p88.

Evidence: moderate • *suggestion (not applied)*

8. With MAOIs (ayahuasca, harmala): without tryptophan, 5-HTP, MDMA or other serotonergics, and without foods rich in tyramine

Pharmacodynamic interactions of MAOIs.

Evidence: strong • *suggestion (not applied)*

9. Do not combine MDMA with oxytocin or excessive water intake

Additive risk of hyponatremia.

Evidence: weak-moderate (mechanism) • *suggestion (not applied)*

10. Esketamine, if it is ever clinically indicated

No adjustment by CYP2B6 (*2/*4). Inform the physician of low dose naltrexone (contradictory data) and telmisartan/tadalafil (vital signals monitored in the session).

Evidence: weak-moderate · *suggestion (not applied)*

Comparison with the example report

The example report is from another person; the comparison shows only where the results differ. Non-identical methods: the percentiles of Sérgio F were calculated with public scores and the reference population.

Trait	Example report	Sérgio F
BDNF	SLIGHTLY LOWER LEVELS	SLIGHTLY LOWER LEVELS
Psychedelic Effects of THC (Functional)	INCREASED	INCREASED
HTR2A (Serotonin)	TYPICAL ACTIVITY	TYPICAL ACTIVITY
CNR1 (Cannabinoids)	TYPICAL ACTIVITY	TYPICAL ACTIVITY
Response to Ketamine	TYPICAL	TYPICAL
CYP2D6 (Mental Health, Detox)	LOWER ACTIVITY	TYPICAL ACTIVITY
CYP2B6 (Drug Metabolism)	TYPICAL ACTIVITY	HIGHER ACTIVITY
CYP2C9 (Detox)	TYPICAL METABOLIZER	INTERMEDIATE METABOLIZER
CYP2C19 (Detox)	EXTENSIVE METABOLIZER	EXTENSIVE METABOLIZER
CYP3A4 (Detox)	TYPICAL ACTIVITY	TYPICAL ACTIVITY
FAAH (Cannabinoids)	LOWER ACTIVITY	LOWER ACTIVITY
Bipolar Disorder	MORE LIKELY (p75)	MORE LIKELY
THC and Psychosis	MORE LIKELY	TYPICAL LIKELIHOOD
Borderline Personality	MORE LIKELY (p93)	MORE LIKELY
Psychological Trauma	MORE LIKELY (p90)	TYPICAL LIKELIHOOD
Non-Epileptic Seizures	MORE LIKELY (p80)	NOT REPRODUCIBLE
Schizophrenia	TYPICAL LIKELIHOOD (p63)	MORE LIKELY
Anxiety	TYPICAL LIKELIHOOD (p57)	TYPICAL LIKELIHOOD
Panic Attacks	TYPICAL LIKELIHOOD (p57)	TYPICAL LIKELIHOOD
Seizures	TYPICAL LIKELIHOOD	TYPICAL LIKELIHOOD
DISC1 (Mental Health)	LOWER ACTIVITY	TYPICAL ACTIVITY
NRG1 (Mental Health)	TYPICAL ACTIVITY	LOWER ACTIVITY

Limitations

- Genotyping chip data plus Beagle imputation and a complete genome reference panel. The SNPs imputed with $DR2 < 0.8$ were treated as not found.
- CYP2D6: it is not possible to see duplications or deletions (CNV, *5, xN) or *3/*6/*9 indels. The probable normal phenotype needs confirmation with a clinical test with CNV.
- MAOA: the promoter's uVNTR, which defines the activity, is not visible; rs6323 is only a marker. UGT1A10 (intestinal route of psilocin) was not evaluated.
- Indels and rare variants are almost never on the chip nor properly impute; rare alleles of CYP2B6, CYP2C9, CYP2C19 or CYP3A4 can go unnoticed.
- PGS explain a small fraction of variance (typically less than 10% of the variance in the risk scale) and are not diagnosed. Family and clinical history weighs more.
- PTSD: coverage of only 34% of the variants of PGS; weak result.
- Borderline (Witt 2017), panic (Forstner 2021) and ketamine (Guo 2018, Li 2020) come from small GWAS, with few variants and without robust replication.
- Non-epileptic seizures: not reproducible; neuroticism is only indicative.
- Counting traits (THC, CNR1, DISC1, NRG1, ketamine) use SNP lists from the example report, with effects of candidate gene studies, often unreplicated.
- The comparison with the reference population (n=633) gives approximate percentiles; the example report uses a reference and its own method.
- The percentiles do not translate into absolute risk. For all evaluated conditions, the absolute risk remains low.

Methods and sources

List of SNPs extracted from the example report (22 traits, 349 lines, 330 rsIDs), plus extra pharmacogenetic 61 SNPs and context.

Genotypes: chip per rsID, then by position (fused rsIDs), then imputation ($DR2 \geq 0.8$). The alleles are in the strand + of dbSNP (GRCh38), confirmed against Ensembl; 0 chain disagreements in the SNPs of the chip.

Scores: PGS of the PGS Catalog (harmonized for GRCh38) and counting scores or weighted from the GWAS Catalog. Percentages calculated against the reference population ($n=633$), with the same SNPs and the same quality rule. Threshold: ≥ 75 MORE LIKELY, ≤ 25 LESS LIKELY.

Pharmacogenetics: diplotypes according to CPIC/PharmVar tables; implications according to CPIC/DPWG when they exist and, in the remaining cases, the indicated literature.

Main sources

- PGS Catalog (Lambert et al., Nat Genet 2021; 2024): PGS002785 (schizophrenia), PGS002786 (bipolar disease), PGS005393 (PTSD), PGS004451 (F41 anxiety/panic), PGS018941 (generalized anxiety), PGS004881 (epilepsy), PGS003565 (neuroticism, proxy).
- GWAS Catalog (Sollis et al., Nucleic Acids Res 2023), associations by rsID and trait.
- International genome reference panel, 3 phase in GRCh38 (reference population, $n=633$) and imputation panel.
- Beagle 5 (Browning et al., Am J Hum Genet 2018; 2021: imputation and $DR2$).
- Ensembl REST / dbSNP: GRCh38 positions and alleles in the strand +.
- CPIC guidelines (CYP2D6, CYP2C9, CYP2C19, CYP2B6, CYP3A5) and PharmVar: definitions of alleles and phenotypes. DPWG for complementary recommendations.
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Generated in 02-10-2026 (local time). The raw genetic data do not leave the local computer.